NATIONAL UNIVERSITY OF MODERN LANGUAGES

Software Construction And Development lab Report - 11

NAME: M Abdul Rafay (FL23791)

M Usman Tanveer (FL23848)

Zulkifal (SP23660)

Elisha Abbasi (SP23606)

PROGRAM: BSSE (5th Semester)

COURSE: Software Construction and Development

SUBMITTED TO: Sir Ahsan Arif

DATE: 9/21/2025 **DAY: Sunday**

LAB 11: Extract Method

Objective

Apply the Extract Method refactoring technique to improve code structure and reduce duplication.

Original Code (Given)

```
public class PayrollCalculator {

  public static void main(String[] args) {


    String employeeName1 = "John";

    double hourlyWage1 = 15.0;

    int hoursWorked1 = 45;


    System.out.println("Employee: " + employeeName1);

    System.out.println("Monthly Salary: $" + hourlyWage * hoursWorked);

    System.out.println();


    String employeeName2 = "Alice";

    double hourlyWage2 = 20.0;

    int hoursWorked2 = 35;


    System.out.println("Employee: " + employeeName2);

    System.out.println("Monthly Salary: $" + hourlyWage * hoursWorked);

    System.out.println();

  }

}
```

Identified Code Smell

- **Code Duplication**:
  The block that prints the employee name and salary appears twice.
- This violates clean code principles and is a perfect case for **Extract Method**.

Refactored Code Using Extract Method:

```
public class PayrollCalculator {

  public static void main(String[] args) {

    printSalary("John", 15.0, 45);
    printSalary("Alice", 20.0, 35);
  }

  private static void printSalary(String name, double wage, int hours) {
    double salary = wage * hours;
    System.out.println("Employee: " + name);
    System.out.println("Monthly Salary: $" + salary);
    System.out.println();
  }
}
```